

Social Engineering — Phishing Awareness Simulation

5.1 Objective

The objective of this exercise was to demonstrate a controlled social-engineering scenario through a phishing-awareness simulation.

The simulation was designed to demonstrate how a user might encounter an account-verification request containing common phishing indicators. The exercise was conducted locally on the Kali Linux system and was designed strictly for cybersecurity awareness and training.

No real credentials were collected, stored, or transmitted during the exercise.

5.2 Simulation Setup

A local HTML-based phishing-awareness simulation was created on the Kali Linux system.

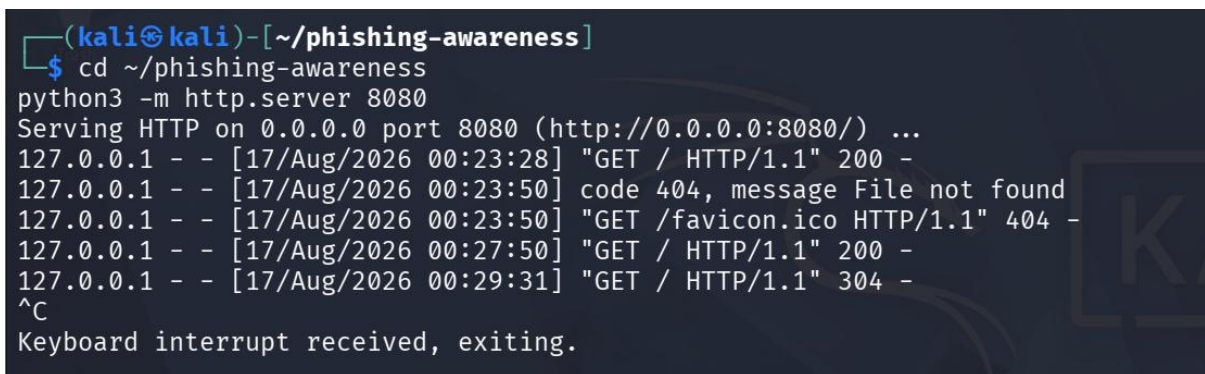
The simulation was hosted using Python's built-in HTTP server:

```
python3 -m http.server 8080
```

The page was accessed locally using:

```
http://127.0.0.1:8080
```

The local server successfully served the webpage, as demonstrated by HTTP 200 responses in the terminal.



```
(kali㉿kali)-[~/phishing-awareness]
$ cd ~/phishing-awareness
$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
127.0.0.1 - - [17/Aug/2026 00:23:28] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [17/Aug/2026 00:23:50] code 404, message File not found
127.0.0.1 - - [17/Aug/2026 00:23:50] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [17/Aug/2026 00:27:50] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [17/Aug/2026 00:29:31] "GET / HTTP/1.1" 304 -
^C
Keyboard interrupt received, exiting.
```

Figure 1: Local HTTP server hosting the phishing-awareness simulation

5.3 Phishing Simulation Page

The simulated webpage presented an **Account Verification** scenario containing an email address field, password field, and verification button.

The page deliberately used an urgent message requesting immediate account verification to demonstrate a common social-engineering technique.

The simulation was clearly labelled:

SECURITY AWARENESS TRAINING — SIMULATION ONLY

No real organization or user's credentials were targeted.

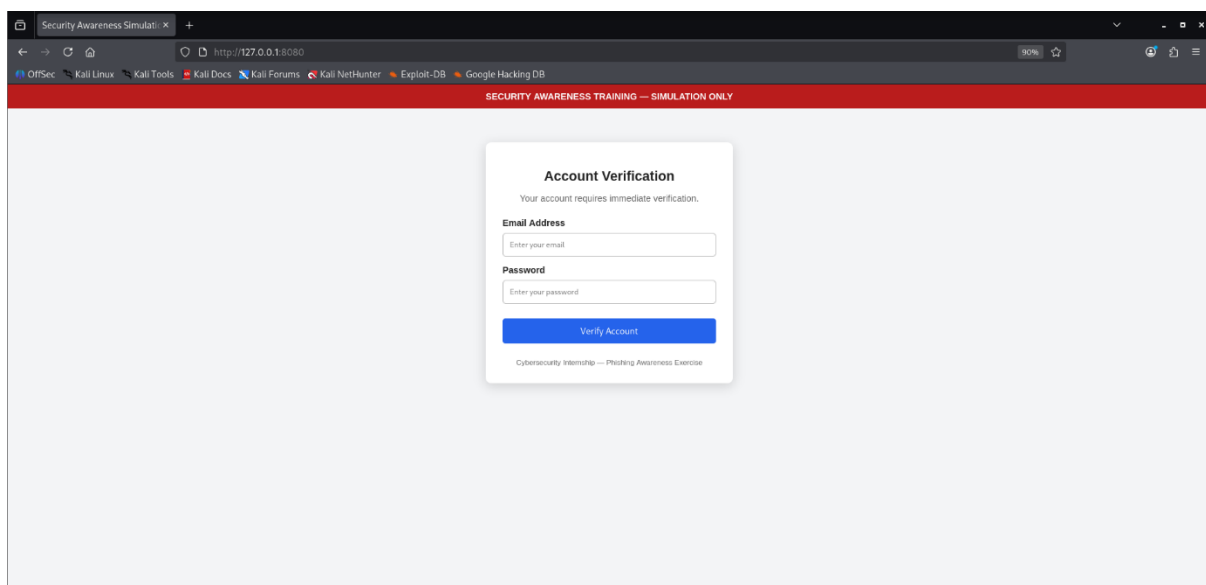


Figure 2: Account-verification phishing-awareness simulation

5.4 User Interaction and Awareness Response

After the simulated form was submitted, the page displayed a **Phishing Awareness Alert**.

The alert explicitly informed the participant that the exercise was a cybersecurity awareness simulation and that:

- No credentials were collected.
- No credentials were stored.
- No credentials were transmitted.

The page then identified several potential phishing indicators.

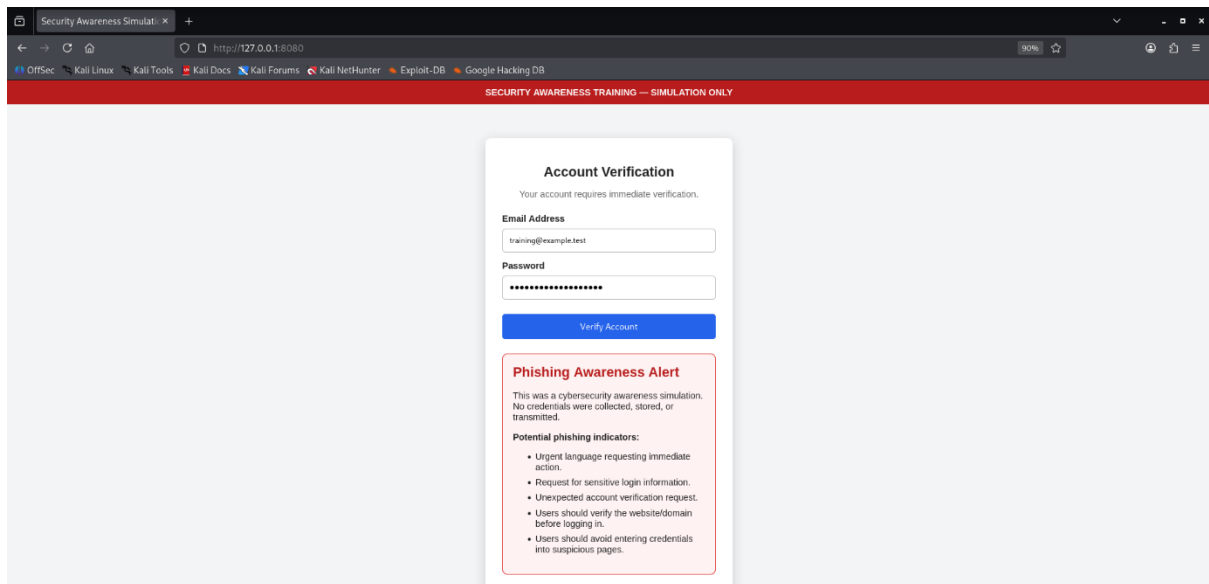


Figure 3: Phishing-awareness alert displayed after simulated interaction

5.5 Phishing Indicators Demonstrated

The simulation highlighted the following indicators:

1. Urgent language

The message stated that the account required **immediate verification**.

Attackers commonly use urgency to encourage users to act without carefully examining a request.

2. Request for sensitive information

The simulated page requested an email address and password.

Users should be cautious whenever an unexpected page requests authentication information.

3. Unexpected account-verification request

An unexpected request to verify an account should be independently checked before the user provides information.

4. Website/domain verification

Users should verify the website address and domain before entering credentials.

5. Avoid entering credentials into suspicious pages

Users should not enter passwords into pages whose authenticity they cannot verify.

5.6 Security Awareness Recommendations

Users should follow these practices when dealing with potential phishing attempts:

- Check the sender and website domain carefully.
- Do not immediately respond to urgent requests.
- Avoid clicking suspicious links.
- Verify account-related requests through an official channel.
- Never provide passwords in response to unexpected requests.
- Use MFA wherever possible.
- Report suspicious messages to the appropriate security or IT team.
- When in doubt, navigate to the organization's official website manually rather than using a link contained in a suspicious message.

5.7 Result

The phishing-awareness simulation was successfully implemented and tested locally.

The exercise demonstrated a complete awareness-training flow:

```
Simulation Page
  ↓
User Interaction
  ↓
Phishing Awareness Alert
  ↓
Identification of Phishing Indicators
```

The exercise did not collect or transmit real credentials and was limited to the local laboratory environment.

5.8 Conclusion

The social-engineering exercise successfully demonstrated the concept of phishing through a controlled awareness simulation.

The simulation showed how urgency, requests for sensitive information, unexpected account verification, and suspicious website characteristics can be indicators of a phishing attempt.

The exercise also demonstrated the importance of user awareness and verification before entering sensitive information.

Because the exercise was designed as a simulation, no real credentials were harvested or transmitted.